

SUJOY GARAI

Backend Engineer | FastAPI · Python · Microservices · REST APIs · System Design · RAG Architecture
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WB

PROFESSIONAL SUMMARY

Backend-focused CS undergraduate with production experience designing and deploying distributed full-stack systems. Proficient in Python, FastAPI, and microservice architecture with hands-on experience integrating cloud services including Firebase, ImageKit, Cloudinary, Docker, and caching layers. Strong understanding of system design, database engineering (PostgreSQL, MongoDB), and multi-platform deployment. Fast learner who continuously explores new technologies and applies them in real production projects. Seeking a Backend Engineering Internship at a product-driven company.

TECHNICAL SKILLS

Languages: Python (Advanced), JavaScript (Intermediate), C (Intermediate), Java (Basic)

Backend: FastAPI, Django, REST APIs, Microservices, JWT Auth, OAuth2, WebSockets, Pydantic

Frontend: React, Vite, Tailwind CSS (Advanced), HTML5, CSS3, Bootstrap

Databases: PostgreSQL, MongoDB, Firebase Realtime DB — schema design, query optimization, indexing

Auth & Storage: Firebase Auth, Firebase Storage, ImageKit, Cloudinary

AI / RAG: RAG Architecture, Chroma Vector DB, Jina Embeddings, LLM Integration, Prompt Engineering

DevOps & Infra: Docker, Load Balancers, Caching, Git, Linux, CI/CD basics

Deployment: Vercel, Render, Railway — production deployments with real user traffic

System Design: Microservice separation, scalability patterns, API gateway, distributed data flow

PROJECTS

Full-Stack RAG System (Production)

React · Vite · Tailwind CSS · FastAPI · Chroma · Jina · PostgreSQL · MongoDB · Firebase · Docker

- Architected a distributed RAG system with React/Vite frontend and a dedicated FastAPI microservice for AI inference — fully decoupled for independent scaling.
- Built document ingestion pipeline with chunking and Jina embeddings; integrated Chroma vector DB for sub-100ms semantic retrieval across concurrent multi-user sessions.
- Implemented Firebase Auth, Firebase Storage, PostgreSQL for relational data, and MongoDB for unstructured metadata across service boundaries.
- Containerized all services with Docker; deployed on Render/Railway with load balancing and caching for production-grade reliability.

Scalable Problem-Solving Platform (Live)

Django · PostgreSQL · JWT Auth · Cloudinary · AI Integration · Vercel/Render

- Built Django REST backend with JWT authentication, role-based access control, and optimized PostgreSQL schema for voting, ranking, and file management.
- Integrated Cloudinary for media storage and AI-based explanation feature with prompt engineering; deployed to production with real user traffic.

EDUCATION

B.Tech, Computer Science & Engineering | Dr. B. C. Roy Engineering College | 2023 – 2027

Class XII (Science) — 90% | Sonamukhi B.J. High School

ACHIEVEMENTS & LEADERSHIP

- Core Member – Google Developer Group (GDG) Durgapur, Web Team — contributed to developer community events and workshops.
- Continuous learner: independently studies system design, cloud infra, and AI/RAG — immediately applies new concepts in live production projects.
- Deployed multiple production systems as an undergraduate with real users, demonstrating full end-to-end backend engineering ownership.